

Regime-Switching Equity Portfolio with Adaptive Allocation

A Deep Reinforcement Learning Approach to Intraday Trading in Indian Markets

Our Solution: “Hybrid Intelligence” Architecture

Three-Tiered System

1. Stock Selection Engine

- **Universe:** Nifty 50 + Nifty Next 50
- **Multi-factor scoring** combining 6 factors:
 - Long-term Momentum (30-day returns)
 - Short-term Momentum (20-day returns)
 - Realized Volatility
 - Average True Range (ATR)
 - Liquidity (dollar volume)
 - Market Beta
- Top 10 stocks selected daily

2. Regime Detection Framework

- **Hidden Markov Model (HMM):** 3-state probabilistic model capturing market regimes (Low Vol, Normal, High Vol)
 - Observations: Log returns, Garman-Klass volatility, Volume Z-scores
- **XGBoost Classifier:** Predicts future regime (Bull/Bear/Sideways) using 50+ features
 - CUSUM detectors, multi-timeframe momentum, EGARCH volatility
 - Technical indicators: RSI, MACD, CCI, CMO
 - EMA crosses and trend structure

3. Reinforcement Learning Trading Agent (PPO)

- **State Space (200+ features):** RSI, CCI, ATR, Heikin-Ashi smoothing, Johnny Ribbon regimes, volume microstructure, Hurst exponent
- **Action Space:** Hold/Exit, Buy, Sell
- **Custom Asymmetric Reward Function:**
 - Linear rewards for profits (encourages upside capture)
 - Quadratic penalties for losses (4x multiplier—agent learns autonomous stop-loss)
 - Structural penalties: Stop-loss triggers, trailing stops, forced EOD closes, missed opportunities

Key Innovations

1. **Walk-Forward Validation:** Rolling 50/50 train-test split prevents overfitting
2. **Sortino-Style Rewards:** Separates “good volatility” (breakouts) from “bad volatility” (crashes)
3. **Trend Structure Veto:** XGBoost filters counter-trend signals during structural downtrends
4. **Risk-Parity Allocation:** Prevents sector concentration
5. **Multi-Layered Risk Controls:** Fixed stops, trailing stops, EOD forced closes

Risk Management

- Training: Stop-loss -5 bps, Trailing stop 5 bps
- Testing: Stop-loss -4 bps, Trailing stop 4 bps
- Forced end-of-day closing to eliminate overnight risk